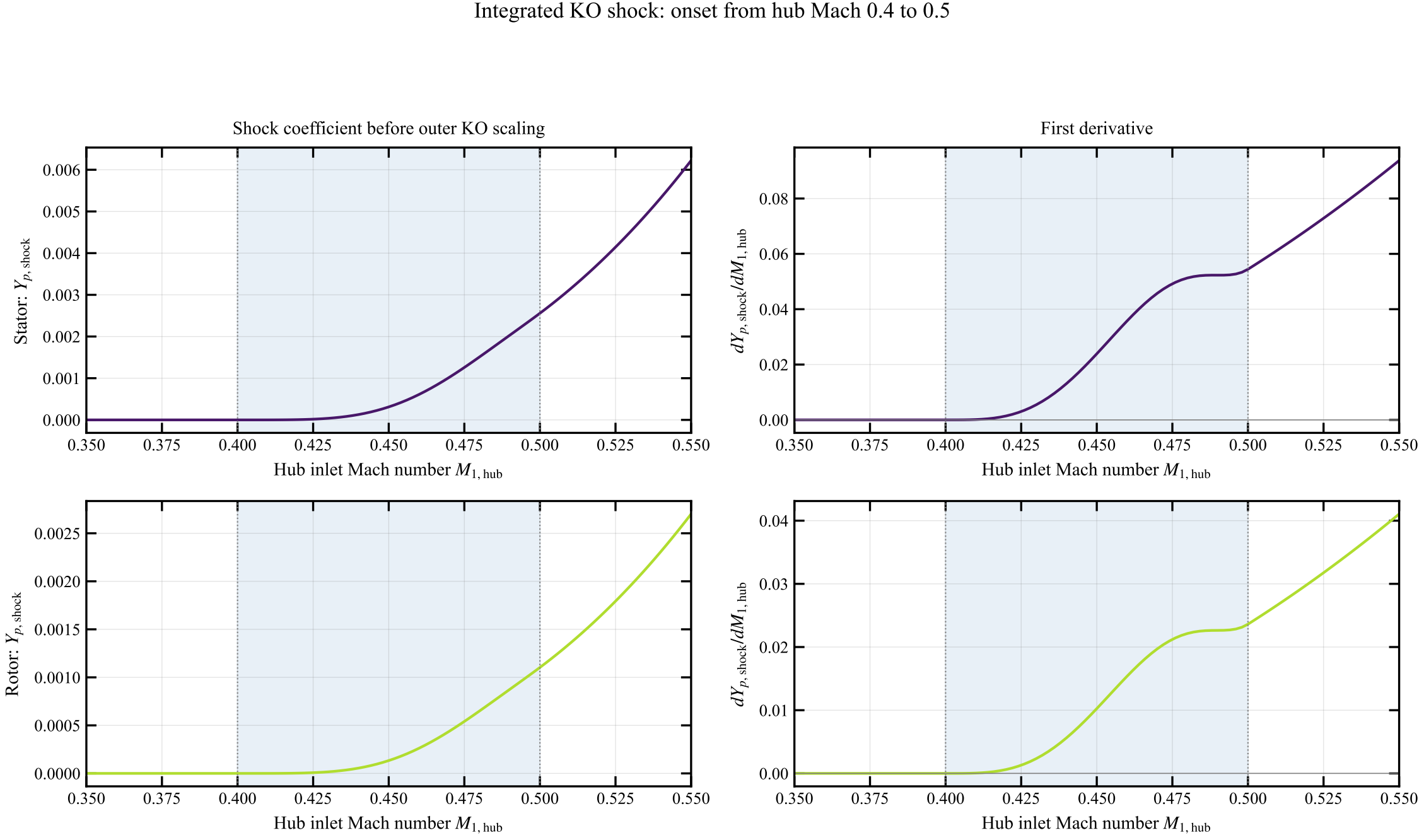

r_h/r_t = 0.5; M2 = 0.6; equal p0 = 200 kPa; gamma = 1.4. Blue: numerical blend interval.
Values and first two derivatives match at both joins; the source helper supplies all plotted components.